Implementation of the Separation Filters: HPSS, Original REPET and REPET-SIM

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November 2026

Abstract

This document describes how the three time–frequency filters of the equalizer are implemented in MATLAB, equation by equation, and how each implementation relates to the paper it comes from: median-filtering harmonic/percussive separation (HPSS) by FitzGerald [1], and the original REPET and REPET-SIM by Rafii, Liutkus and Pardo [2, 3, 4]. It covers the shared STFT front end, the meaning of each user parameter and its conversion from physical units, the resulting masks, the way the masks are combined into a musical component, and every point where the code deliberately departs from the papers or from the authors' reference code. The gain stage and the energy measurement are out of scope and will be documented separately.

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1 Scope and notation

The equalizer boosts or attenuates a *musical component* (e.g. the lead voice, the percussion, the harmonic accompaniment) of a stereo song. A component is not separated as a stand-alone track; it is described by a time–frequency mask $M \in [0, 1]$ that weights every STFT bin of every channel. Two independent filters provide the information used to build that mask:

- **Character**: HPSS classifies each bin as *sustained* (harmonic) or *percussive*.
- **Repetition**: either the original REPET or REPET-SIM (selected globally in the interface) classifies each bin as part of a *repeating* background or of a *non-repeating* foreground.

None of the filters uses machine learning or pre-trained models: all three are built from the STFT, median filters and element-wise operations, and all of them are implemented from scratch without any MATLAB toolbox.

Notation. Throughout the document:

Symbol	Meaning
$x_c[\nu]$	channel $c \in \{1, \ldots, C\}$ of the fragment, $C = 1$ or 2, sampling rate f_s
N_w, R	window/FFT length and hop size [samples]
$X_c(i, j)$	complex STFT of channel c, bin $i = 1, \ldots, n$ ($n = N_w/2 + 1$), frame $j = 1, \ldots, m$
$V_c(i, j)$	magnitude spectrogram $\lvert X_c(i, j)\rvert$ (the `F.Vch` array, size $n \times m \times C$)
$\bar{V}(i, j)$	channel average $\frac{1}{C}\sum_c V_c(i, j)$ (`mean(V,3)`)
$\Delta f = f_s/N_w$	bin spacing [Hz]; $\Delta t = R/f_s$ frame spacing [s]
M_H, M_P	HPSS sustained and percussive masks, $M_P = 1 - M_H$
M_R	REPET background (repeating) mask; $1 - M_R$ is the foreground

FitzGerald indexes bins by h and frames by i; the REPET chapter uses i for bins and j for frames. This document follows the REPET convention (i = bin, j = frame) for all three filters. MATLAB identifiers are quoted as they appear in the code (in Spanish).

Source files.

File	Role
`core/ventana_hann.m`	periodic Hann window
`core/calcular_stft.m`, `core/calcular_istft.m`	STFT and WOLA inverse
`core/caracteristicas_tf.m`	STFT of each channel and magnitudes, once per fragment
`core/hpss_mascara.m`	HPSS mask, FitzGerald (2010)
`core/repet_mascara.m`	original REPET mask, Rafii et al. (2014) §14.2
`core/repet_sim_mascara.m`	REPET-SIM mask, Rafii et al. (2014) §14.4
`core/rampa_coseno.m`	raised-cosine band/pan edges
`core/mascara_componente.m`	product of the masks that defines a component
`core/aplicar_componentes.m`	applies the component masks and inverts (gain stage)

2 Shared time–frequency front end

All three filters operate on the *same* STFT. This is not a convenience: the masks are multiplied bin by bin (section 3), so they must live on the same time–frequency grid.

2.1 Analysis window

`ventana_hann` returns the *periodic* Hann window

$$w[\nu] = \tfrac{1}{2} - \tfrac{1}{2}\cos\!\left(\frac{2\pi\nu}{N_w}\right), \qquad \nu = 0, \ldots, N_w - 1. \tag{1}$$

The periodic (rather than symmetric) form is used because shifted copies of it at hops $R = N_w/2$ or $N_w/4$ add up to a constant, which keeps the normalisation of the inverse transform well conditioned.

2.2 Forward STFT (`calcular_stft`)

1. The signal is zero-padded with $N_w/2$ samples at the start and $N_w/2 + N_w$ at the end, so that frame j is *centred* on sample $(j-1)R$ of the original signal.
2. The number of frames is $m = \lceil (N-1)/R \rceil + 1$, so the last frame is centred at or after the last sample and every sample is covered by frames on both sides.
3. Each frame is multiplied by w, transformed with an N_w-point FFT, and only the non-negative frequencies are kept: $n = N_w/2 + 1$ bins at $f_i = (i-1)\Delta f$.

The frames are built at once with an index matrix (`idx = (1:Nw)' + (0:nTramas-1)*hop`), so the whole STFT is a single vectorised FFT call.

2.3 Inverse STFT (`calcular_istft`)

The inverse is weighted overlap–add (WOLA): every frame is rebuilt from its Hermitian-symmetric spectrum, inverse-transformed (`'symmetric'` flag, so the output is exactly real), multiplied again by the synthesis window and added at its position; the sum is divided by the overlap of the squared windows:

$$y[\nu] = \frac{\sum_j w[\nu - (j-1)R]\; \mathcal{F}^{-1}\{Y_j\}[\nu - (j-1)R]}{\sum_j w^2[\nu - (j-1)R]}. \tag{2}$$

If $Y = X$ (no modification) this returns x exactly (round-off of order 10^{-15}). If Y has been masked, eq. (2) is the Griffin–Lim least-squares estimate [5]: the signal whose STFT is closest to the modified one. Two properties are used later:

- **Linearity**: $\mathrm{ISTFT}(aY_1+bY_2) = a\,\mathrm{ISTFT}(Y_1)+b\,\mathrm{ISTFT}(Y_2)$. Hence the REPET foreground, which the paper obtains as "mixture minus background" in the time domain, is identical to applying the mask $1 - M_R$.
- **Exact reconstruction**: masks that add up to one in every bin reconstruct the original signal exactly.

2.4 Default parameters

The default STFT is a Hann window of $N_w = 4096$ samples with 75 % overlap ($R = 1024$), which is the setting used by FitzGerald [1] (FFT 4096, hop 1024 at 44.1 kHz), who also reports better separation with longer FFTs. The REPET papers do not prescribe a window (see section 8). Window and overlap can be changed in the interface panel *"Análisis STFT (afecta el procesamiento)"*; all filter parameters are entered in seconds or hertz and re-converted, so their physical meaning does not change.

Table 1: Resolution of the default STFT (Hann 4096, hop 1024).

Quantity	$f_s = 48$ kHz	$f_s = 44.1$ kHz
Bin spacing $\Delta f = f_s/N_w$	11.7 Hz	10.8 Hz
Window duration N_w/f_s	85.3 ms	92.9 ms
Hop $\Delta t = R/f_s$	21.3 ms	23.2 ms
Hann main lobe ($4\Delta f$)	46.9 Hz	43.1 Hz
Frames in a 20 s fragment, m	939	863

2.5 Per-fragment features (`caracteristicas_tf`)

For a fragment of size $N \times C$ this function computes, once: the STFT X_c of each channel (stored in single precision), the magnitudes $V_c = |X_c|$ stacked as an $n \times m \times C$ array (`F.Vch`, the input of all three filters), the mono power $|(X_1 + X_2)/2|^2$ (for display), and a panning index $p(i, j) = (|X_R| - |X_L|)/(|X_R| + |X_L|) \in [-1, 1]$. The HPSS and REPET masks are computed afterwards, so that their parameters can be changed without recomputing the STFT.

3 Common masking framework

3.1 Per-channel masks

Each filter receives V ($n \times m \times C$) and returns a mask of the same size, i.e. *one mask per channel*. Where a filter needs a single decision for both channels (the REPET period and the REPET-SIM similarity matrix), that decision is taken on the channel average $\bar{V}$, and the mask itself is then built from each V_c separately, as in the authors' reference code.

3.2 Complementary pairs

Each filter produces a mask and its complement:

$$M_P = 1 - M_H, \qquad M_{\mathrm{fg}} = 1 - M_R. \tag{3}$$

HPSS gives a *binary* mask ($M_H \in \{0, 1\}$, FitzGerald eqs. 8–9), both REPET variants give a *soft* mask ($M_R \in [0, 1]$, Rafii eq. 14.4). The masks are used exactly as the papers define them: no thresholding, sharpening, or frequency cut-off is applied inside the filters.

3.3 Component mask (`mascara_componente`)

A component is the product of four independent cues:

$$M(i,j,c) = M_{\text{band}}(i) \cdot M_{\text{char}}(i,j,c) \cdot M_{\text{rep}}(i,j,c) \cdot M_{\text{pan}}(i,j), \tag{4}$$

where

- M_{band} is 1 inside $[f_1, f_2]$ with raised-cosine edges of width `suavF` (`rampa_coseno`);
- $M_{\text{char}} \in \{1,\ M_H,\ 1 - M_H\}$ for *cualquiera / sostenido / percusivo*;
- $M_{\text{rep}} \in \{1,\ M_R,\ 1 - M_R\}$ for *cualquiera / repetitivo / norepetitivo*;
- M_{pan} selects a panning range with cosine edges; it is only applied when the fragment is truly stereo (inter-channel correlation < 0.99).

The cosine edges matter: multiplying the spectrum by a rectangular band is a convolution with a sinc in time and produces audible ringing.

3.4 Partition of unity

Because every cue comes as a complementary pair, the four character $\times$ repetition combinations add up to one in every bin:

$$M_H M_R + M_H(1 - M_R) + (1 - M_H)M_R + (1 - M_H)(1 - M_R) = 1. \tag{5}$$

By the exact-reconstruction property of the WOLA inverse, the four components played together give back the original signal. This is an invariant of the design and is checked by the tests (section 9). The typical lead-voice component is *Sostenido* $\times$ *No repetitivo*, i.e. $M = M_H(1 - M_R)$: a product of a binary and a soft mask.

3.5 Interface with the gain stage

`aplicar_componentes` consumes these masks: the processed STFT of channel c is $Y_c = G(\cdot,\cdot,c) \odot X_c$ with a per-bin gain built from the component masks, and $y_c = \text{ISTFT}(Y_c)$. The design of the gain law and its energy measurement are outside the scope of this document.

4 HPSS: harmonic/percussive separation by median filtering

Reference: FitzGerald (2010) [1]. Code: `core/hpss_mascara.m`.

4.1 Principle

In a spectrogram, a stationary pitched sound (a held note, a vowel) is a *horizontal* ridge: energy that stays in the same bin across many frames. A percussive onset is a *vertical* ridge: energy spread over many bins during very few frames. FitzGerald's observation is that each kind of event is an *outlier* in the direction in which the other one is smooth:

- along *time*, in a fixed bin, a drum hit is a short spike on top of the slowly-varying harmonic energy;
- along *frequency*, in a fixed frame, a harmonic is a narrow spike on top of the broadband percussive energy.

A median filter removes outliers that occupy less than half of its window, so median filtering in each direction suppresses one class and keeps the other. This replaces the iterative diffusion of Ono et al. with two non-iterative passes.

4.2 Equations

Median filter (eq. 5). For odd length l,

$$y(\nu) = \text{median}\{x(\nu - k), \ldots, x(\nu + k)\}, \qquad k = \frac{l-1}{2}. \tag{6}$$

Percussion-enhanced spectrogram (eq. 6). Each frame j is median-filtered *across frequency* with length l_{perc}:

$$P(:, j) = \mathcal{M}\{V(:, j),\, l_{\text{perc}}\}. \tag{7}$$

Harmonic-enhanced spectrogram (eq. 7). Each bin i is median-filtered *across time* with length l_{harm}:

$$H(i, :) = \mathcal{M}\{V(i, :),\, l_{\text{harm}}\}. \tag{8}$$

Binary masks (eqs. 8–9).

$$M_H(i,j) = \begin{cases} 1 & H(i,j) > P(i,j) \\ 0 & \text{otherwise} \end{cases}, \qquad M_P(i,j) = \begin{cases} 1 & P(i,j) > H(i,j) \\ 0 & \text{otherwise} \end{cases}. \tag{9}$$

Soft (Wiener) masks (eqs. 10–11), not implemented yet.

$$M_H = \frac{H^p}{H^p + P^p}, \qquad M_P = \frac{P^p}{H^p + P^p}, \qquad p \in \{1, 2\}. \tag{10}$$

Resynthesis (eqs. 12–13). The masks multiply the original *complex* STFT, $\hat{H} = X \odot M_H$, $\hat{P} = X \odot M_P$, which are then inverted.

4.3 Implementation

The function signature is

```
[Marm, Lt, Lf] = hpss_mascara(V, fs, Nw, hop, tArm, fPerc, tipo)
```

Algorithm 1 HPSS mask (`hpss_mascara`)

Require: $V \in \mathbb{R}_{\geq 0}^{n \times m \times C}$, f_s, N_w, R, t_{arm} [s], f_{perc} [Hz]

1: $L_t \leftarrow \text{odd}(\max(3,\ \text{round}(t_{\text{arm}} f_s / R)))$ ▷ l_{harm} in frames
2: $L_f \leftarrow \text{odd}(\max(3,\ \text{round}(f_{\text{perc}} N_w / f_s)))$ ▷ l_{perc} in bins
3: $H \leftarrow$ `movmedian(V, Lt, 2)` ▷ along time, every bin and channel (eq. 7)
4: $P \leftarrow$ `movmedian(V, Lf, 1)` ▷ along frequency, every frame and channel (eq. 6)
5: $M_H \leftarrow [H > P]$ (single, values 0/1) ▷ eq. 8
6: **return** M_H ▷ $M_P = 1 - M_H$ is formed in `mascara_componente`

Here $\text{odd}(l)$ adds one when l is even, so that the filter has a well-defined centre sample as eq. 5 requires. Because `movmedian` operates along one dimension of the full 3-D array, both channels are processed in the same call and each channel gets its own mask.

4.4 Parameters

The paper uses 17×17 filters at FFT 4096, hop 1024, 44.1 kHz, and reports that separation quality "did not vary dramatically" for lengths between 15 and 30. The code takes the lengths in physical units, chosen to reproduce 17×17 at the paper's setting:

$$17 \text{ frames} \times 23.2 \text{ ms} \approx 400 \text{ ms}, \qquad 17 \text{ bins} \times 10.8 \text{ Hz} \approx 180 \text{ Hz}. \tag{11}$$

Table 2: HPSS parameters (default STFT).

Parameter	Default	48 kHz	44.1 kHz	UI range
Time median l_{harm} (`tArm`)	400 ms	19 frames (405 ms)	17 frames (395 ms)	100–1500 ms
Freq. median l_{perc} (`fPerc`)	180 Hz	15 bins (176 Hz)	17 bins (183 Hz)	50–1000 Hz

What each parameter does. A median keeps whatever occupies more than half of its window. Therefore:

- l_{harm} is roughly *twice the minimum duration* an event must have to be called sustained. Increasing it (e.g. 800 ms) makes only long notes sustained: short syllables, fast melodic passages and vibrato move to the percussive side. Decreasing it (e.g. 200 ms) lets cymbal tails and resonant hits become sustained, so percussion leaks into the sustained component.
- l_{perc} is roughly *twice the minimum bandwidth* an event must have to be called percussive. Increasing it gives a smoother percussive estimate and pushes broad partials (chorus, vibrato) towards the sustained side; decreasing it lets wide harmonic lobes or dense chords be called percussive.

For the sensitivity sweep of the report, the paper's stable range 15–30 corresponds at 48 kHz to approximately 320–640 ms and 176–352 Hz.

4.5 Implementation decisions relative to the paper

1. **Ties.** Eqs. 8–9 leave bins with $H = P$ in *neither* class ($M_H = M_P = 0$). In practice ties happen in digital silence ($H = P = 0$). The code defines $M_P = 1 - M_H$, which sends ties to the percussive side and keeps the partition of unity (eq. (5)) exact.
2. **Borders.** The paper does not say how the median behaves at the edges of the spectrogram. `movmedian` uses its default `'shrink'` mode: near the first/last frame and near 0 Hz and Nyquist the window is truncated to the available samples.
3. **Stereo.** The paper treats mono signals and notes the method "can easily be extended to handle stereo". Here it is applied to each channel independently.
4. **Binary mask.** The paper observes that the soft mask with $p = 2$ produces fewer artefacts at the cost of slightly more interference, and that $p = 2$ is considerably better than $p = 1$. The current code implements only the binary mask; the argument `tipo` is reserved for the soft variant.

4.6 Expected behaviour and limitations

- **Musical noise.** A binary mask switches bins fully on or off, so isolated bins flip between classes from frame to frame. With large gains (±18–20 dB) this becomes audible as "musical noise" and hard spectral edges.
- **Low notes.** With $N_w = 4096$ at 48 kHz the Hann main lobe of a single partial spans about 4 bins (≈ 47 Hz). When the fundamental is below about 95 Hz (bass notes E1–G2), consecutive harmonics are closer than two main lobes and occupy more than half of the 15-bin frequency window, so part of a sustained bass note is classified as percussive.
- **Attacks.** Note onsets of plucked or struck instruments (guitar, piano, bass) are broadband and short: they are correctly, but often undesirably, assigned to the percussive side, as the paper itself reports.

- **Pitch glides.** Sung vibrato and slides are not horizontal; energy that moves by more than a bin within $l_{\text{harm}}/2$ leaks to the percussive side.

Cost: two moving medians over the spectrogram, $O(nm\,(l_{\text{harm}} + l_{\text{perc}}))$ in the worst case; it is the cheapest of the three filters.

5 Original REPET: repetition with a single period

References: Rafii and Pardo (2013) [3]; Rafii, Liutkus and Pardo (2014), §14.2 [2]. Code: `core/repet_mascara.m`.

5.1 Principle

Popular music is frequently a varying foreground (the voice) laid over a background that repeats with a fixed period (a bar, a drum loop, a chord cycle). If that period p is known, the spectrogram can be cut into segments of p frames; at corresponding positions the background is the same in every segment, while the voice appears in only some of them. The element-wise median across segments therefore keeps the background and rejects the voice as an outlier. The method has three stages: (1) find the period, (2) model a repeating segment, (3) extract the repeating structure with a soft mask.

5.2 Equations

Beat spectrum (eq. 14.1). With V^2 the power spectrogram (averaged over channels for stereo),

$$A(i,l) = \frac{1}{m-l+1}\sum_{j=1}^{m-l+1} V(i,j)^2\,V(i,j+l-1)^2, \qquad b(l) = \frac{1}{n}\sum_{i=1}^{n} A(i,l), \qquad b(l) \leftarrow \frac{b(l)}{b(1)}. \quad (12)$$

A is the *unbiased* autocorrelation over time of each frequency row of V^2; b is its mean over frequency. The squaring emphasises the periodicity peaks. Lag $l = 1$ in the paper's 1-based notation is lag 0.

Period. The paper accepts any "period finder"; the reference code takes the maximum of b inside a given lag range.

Repeating segment model (eq. 14.2). With r segments of length p,

$$S(i,j) = \underset{k=1\ldots r}{\text{median}}\, V\big(i,\; j+(k-1)p\big), \qquad j = 1,\ldots,p. \quad (13)$$

Repeating spectrogram model (eq. 14.3).

$$W\big(i,j+(k-1)p\big) = \min\Big(S(i,j),\; V\big(i,j+(k-1)p\big)\Big). \quad (14)$$

Since V is assumed to be the sum of a non-negative background W and a non-negative foreground $V - W$, the background cannot exceed the mixture.

Soft mask (eq. 14.4).

$$M(i,j) = \frac{W(i,j)}{V(i,j)} \in [0,1]. \quad (15)$$

The paper notes that a binary mask (threshold on M) is possible but that pilot experiments favoured the soft one.

Algorithm 2 Original REPET mask (`repet_mascara`)

Require: $V \in \mathbb{R}^{n\times m\times C}_{\geq 0}$, f_s, R, $[p_{\min}, p_{\max}]$ [s]

$Q \leftarrow \bar{V}^{\,2}$ ▷ `mean(V,3).^2`, $n \times m$

$A \leftarrow \mathcal{F}^{-1}\{|\mathcal{F}_{2m}\{Q(i,:)\}|^2\}$ for every row i ▷ linear autocorrelation via zero-padded FFT

$A(i,l) \leftarrow A(i,l)/(m-l)$ for lags $l = 0, \ldots, m-1$ ▷ unbiased

$b \leftarrow \frac{1}{n}\sum_i A(i,:)$; $\quad b \leftarrow b/(b(0)+\varepsilon)$

$[l_{\text{lo}}, l_{\text{hi}}] \leftarrow \text{round}([p_{\min}, p_{\max}]\, f_s/R)$; $\quad l_{\text{lo}} \leftarrow \max(1, l_{\text{lo}})$; $\quad l_{\text{hi}} \leftarrow \min(l_{\text{hi}}, \lfloor m/3 \rfloor)$

if $l_{\text{hi}} < l_{\text{lo}}$ **then** ▷ fragment too short for 3 periods

  $l_{\text{hi}} \leftarrow \min(\max(l^{\text{orig}}_{\text{hi}}, l_{\text{lo}}),\, m-1)$; $\quad l_{\text{lo}} \leftarrow \min(l_{\text{lo}}, l_{\text{hi}})$

end if

$p \leftarrow \arg\max_{l\in[l_{\text{lo}},l_{\text{hi}}]} b(l)$; $\quad r \leftarrow \lceil m/p \rceil$

for $c = 1, \ldots, C$ **do**

  pad V_c with NaN to rp frames and reshape to $n \times p \times r$

  $S \leftarrow$ `median(.,3,'omitnan')` ▷ eq. 14.2

  $W \leftarrow \min(V_c, S)$ ▷ eq. 14.3, segment by segment

  $M_R(:,:,c) \leftarrow (W+\varepsilon)/(V_c+\varepsilon)$, truncated to m frames ▷ eq. 14.4

end for

return M_R, diagnostics (b, p, $p\Delta t$, r, $60/(p\Delta t)$)

5.3 Implementation

Signature: `[Mrep, info] = repet_mascara(V, fs, hop, rangoPeriodo)`.

Notes on the steps.

- **Autocorrelation via FFT.** By the Wiener–Khinchin theorem, the inverse FFT of $|\mathcal{F}\{q\}|^2$ is the autocorrelation of q. A plain m-point FFT would give the *circular* autocorrelation (lag l would also correlate the end of the fragment with its beginning); zero-padding to $2m$ makes it linear. Dividing lag l by $m-l$ (the number of overlapping terms) is exactly the $1/(m-l+1)$ factor of eq. 14.1 in 0-based lags. This gives the $O(m \log m)$ cost quoted in the chapter.

- **Upper limit $\lfloor m/3 \rfloor$.** As in the reference code, the period cannot exceed a third of the fragment, so that at least three segments enter the median. With fewer than three values a median cannot reject an outlier.

- **Partial last segment.** When m is not a multiple of p, the last segment is incomplete; padding with NaN and using `'omitnan'` makes the median at those positions use only the segments that exist. `min` ignores NaN as well, and the padded columns are discarded at the end.

- **ε in the mask.** ε is MATLAB's `eps` (2.2×10^{-16}), as in the reference code. It avoids 0/0: a silent bin gets $M = 1$ ("background"), which is harmless because $X = 0$ there.

5.4 Parameters

The defaults $[1, 10]$ s are those of the authors' reference code. Note that the $m/3$ limit dominates for the 20 s fragments of the assignment: with $m = 939$ frames at 48 kHz, $\lfloor m/3 \rfloor = 313$ frames $= 6.7$ s, so the effective search range is $[1, 6.7]$ s, not $[1, 10]$ s.

Diagnostics shown in the interface. The selected period in seconds and frames, the number of segments r (including the partial one), and the *equivalent BPM* $60/(p\,\Delta t)$, together with $\times 2$ and $\times 4$, because p is usually a bar (2 or 4 beats) rather than a single beat. A warning is shown

Table 3: Original REPET parameters.

Parameter	Default	UI range	Effect
Minimum period $p_{\min}$	1 s	0.1–30 s	Shortest lag allowed. Raising it skips the beat and selects a bar or a phrase. Lowering it allows the beat or a subdivision; if the voice lasts longer than that period it is copied into the model and treated as background.
Maximum period $p_{\max}$	10 s	0.1–30 s, and $\leq m/3$	Longest lag allowed. Raising it allows long phrases but reduces r. Lowering it forces short periods; if the true pattern is longer, the median mixes different parts of the pattern.

when $r < 3$. The button *"Ver espectro de ritmo"* plots $b(l)$ with the search range shaded and the chosen period and its multiples marked.

5.5 Limitations

- A single period for the whole fragment: tempo changes or a change of section (verse → chorus) break the model. This is what the adaptive REPET (§14.3, not implemented) and REPET-SIM address.
- In the unbiased beat spectrum the multiples $2p, 3p, \ldots$ have peaks of similar height, so the maximum may be a multiple of the musically "right" period. The separation remains valid while $r \geq 3$, but the beat-spectrum plot should be checked.
- A repeated vocal phrase (a chorus sung identically every period) is background by construction.

6 REPET-SIM: repetition through the similarity matrix

References: Rafii and Pardo (2012) [4]; Rafii, Liutkus and Pardo (2014), §14.4 [2]. Code: `core/repet_sim_mascara.m`.

6.1 Principle

REPET-SIM drops the assumption of a period. For every frame j it finds the frames that *look most like it* anywhere in the fragment, and takes the median over them. Background material repeats (not necessarily periodically), so it is present in most of those frames and survives the median; the foreground is present in few of them and is rejected.

Example. A 440 Hz bin over five similar frames has magnitudes $[0.30, 0.31, 0.29, \mathbf{0.90}, 0.30]$; the voice is only in the fourth frame. The median is 0.30, so in that frame the mask is $0.30/0.90 = 0.33$: 33 % background, 67 % foreground.

6.2 Equations

Similarity matrix (eq. 14.8). Cosine similarity between frames:

$$S(j_a, j_b) = \frac{\sum_i V(i, j_a) V(i, j_b)}{\sqrt{\sum_i V(i, j_a)^2}\sqrt{\sum_i V(i, j_b)^2}}, \qquad j_a, j_b = 1, \ldots, m. \tag{16}$$

Repeating frames. For each frame j, the indices J_j of the most similar frames are kept, subject to three parameters: a minimum similarity t, a minimum distance d between two selected frames, and a maximum number k of selected frames.

Initial model (eq. 14.9).

$$U(i,j) = \underset{l=1\ldots k}{\text{median}}\, V\big(i, J_j(l)\big). \tag{17}$$

Refined model (eq. 14.7) and mask (eq. 14.4). $W = \min(U, V)$ and $M = W/V$, exactly as in eq. (15).

6.3 Implementation

Signature: `[Mrep, info] = repet_sim_mascara(V, fs, hop, k, d, t)`.

Algorithm 3 REPET-SIM mask (`repet_sim_mascara`)

Require: $V \in \mathbb{R}_{\geq 0}^{n\times m\times C}$, f_s, R, k, d [s], t
$D \leftarrow \max(1, \text{round}(d\, f_s/R))$ ▷ minimum distance in frames
$\bar{V}(:,j) \leftarrow \bar{V}(:,j)/(\|\bar{V}(:,j)\|_2 + \varepsilon)$ (single)
$S \leftarrow \bar{V}^{\mathsf{T}}\bar{V}$ ▷ $m \times m$ cosine matrix, eq. 14.8
for $j = 1, \ldots, m$ **do**
  $J_j \leftarrow \textsc{Peaks}(S(:,j),\, t,\, D,\, k)$
  if $J_j = \emptyset$ **then** $J_j \leftarrow \{j\}$
  end if
end for
for $c = 1, \ldots, C$ **do**
  $U(:,j) \leftarrow \text{median}\, V_c(:, J_j)$ for every j ▷ eq. 14.9
  $W \leftarrow \min(U, V_c)$ ▷ eq. 14.7
  $M_R(:,:,c) \leftarrow (W + \varepsilon)/(V_c + \varepsilon)$ ▷ eq. 14.4
end for
return M_R and diagnostics

Peak selection (`picos_columna`). The reference code selects the repeating frames with

```
findpeaks(S(:,j), 'MinPeakHeight', t, 'MinPeakDistance', D,
                  'SortStr', 'descend', 'NPeaks', k),
```

which belongs to the Signal Processing Toolbox. Since `core/` must be toolbox-free, the same behaviour is re-implemented:

1. a *peak* is a sample strictly greater than both neighbours; on a plateau only its first sample counts; the first and last samples are never peaks;
2. only peaks with height $> t$ are kept;
3. in descending order of height, a peak is accepted and every remaining candidate closer than D frames to it is discarded;
4. at most k peaks are returned.

Since $S(j,j) = 1$ is the global maximum of column j, the frame itself is the first peak accepted, except for $j = 1$ and $j = m$, which are end points and therefore never peaks (identical to `findpeaks`). This has a useful consequence: its $D-1$ neighbours on each side are then discarded, so d effectively excludes the frames that resemble j only because the same note is still sounding.

Channels. The similarity matrix and the selected indices J_j are computed once on $\bar{V}$ (as in the reference code); the median model and the mask are computed on each channel's own V_c.

6.4 Parameters

Interaction between k and d on short fragments. With $D = 47$ frames ($d = 1$ s at 48 kHz) and $m = 939$ frames (20 s), at most $\lfloor 938/47 \rfloor + 1 = 20$ frames can be selected. Hence $k = 100$

Table 4: REPET-SIM parameters (defaults from the authors' reference code).

Parameter	Default	UI range	Effect
k, max. repeating frames	100	3–500	Number of frames (including j itself) entering the median. More frames give a more robust model *if* that many true repetitions exist; otherwise poorly matching frames enter and the background is over-estimated. With $k < 3$ the median cannot reject anything.
d, min. distance	1 s	0.1–10 s	Minimum separation between two selected frames. Larger d prevents a long held note (including a sung one) from being matched with itself and called repeating, but limits the number of frames to $\lfloor (m-1)/D \rfloor + 1$. Smaller d lets sustained vocals leak into the background.
t, min. similarity	0	0–0.99	Minimum cosine similarity to accept a frame. Raising it (0.6–0.8) keeps only real repetitions; in passages that do not repeat few frames remain, $U \approx V$, $M \approx 1$, and nothing is extracted there. With $t = 0$ any peak is accepted.

never acts on a 20 s fragment: the limit is set by d, and almost every region of the fragment enters every median. k becomes relevant only for full songs.

Diagnostics shown in the interface. Average number of frames per median, the maximum that fits for the chosen d, the fraction of frames for which k was the binding limit, and the mean similarity of the selected frames (excluding j itself). A low mean similarity (≈ 0.5–0.7) means the median is built from frames that are not real repetitions; the remedy is to raise t or to analyse a longer excerpt.

6.5 Cost and limitations

- Time and memory are $O(m^2)$ because of S. A 3-minute song at 48 kHz with $R = 1024$ has $m \approx 8400$ frames, so S alone takes ≈ 280 MB in single precision (the reason S is stored as `single`).
- As in REPET, a repeated vocal phrase is background, and non-repeating instrumental fills fall on the foreground side.

7 Comparison of the two repetition filters

	Original REPET	REPET-SIM
Assumption	background repeats with one fixed period	background repeats, not necessarily periodically
Frames in the median	one per segment, at multiples of p	up to k most similar frames, $\geq d$ apart
Decision taken on	beat spectrum of $\bar{V}^2$	cosine matrix of $\bar{V}$
Cost	$O(m \log m)$	$O(m^2)$
Fails when	tempo or section changes	few true repetitions, or d too small
Diagnostics	beat spectrum, p, r, BPM	frames per median, mean similarity

REPET-SIM does not estimate any period, so it has no BPM; a tempo value would be a separate computation that the filter does not use.

8 Consolidated list of deviations

The guiding rule of the implementation is fidelity to the papers and to the authors' reference code. Every departure is listed here.

Filter	Item	Paper / reference	This implementation and reason
All	STFT	REPET reference: $2^{\lceil \log_2(0.04 f_s) \rceil} = 2048$ samples, 50 % overlap, Hamming. FitzGerald: 4096 / 1024.	Hann 4096, 75 %. One STFT is needed for all filters (masks are multiplied bin by bin); FitzGerald fixes his and reports better results with longer FFTs, the REPET papers do not fix one. The hop (1024) equals the reference's, so the frame rate and all parameters in seconds are unchanged.
HPSS	Ties $H = P$	Neither class	Percussive ($M_P = 1 - M_H$), to keep eq. (5) exact.
HPSS	Median borders	Not specified	`movmedian 'shrink'`.
HPSS	Stereo	Mono only	One mask per channel.
HPSS	Mask	Soft $p = 2$ recommended	Binary only (soft pending).
REPET / SIM	100 Hz cut-off	Reference code forces $M = 1$ below 100 Hz	Removed: not part of the papers. Low frequencies are classified by repetition only; a band limit on the component is the explicit alternative.
REPET / SIM	Mask ε	$M = W/V$	$(W + \varepsilon)/(V + \varepsilon)$, as in the reference code.
REPET	Stereo beat spectrum	Chapter text: average V^2 over channels	$(\bar{V})^2$, as the reference code does. Note that $(\bar{V})^2 \neq \overline{V^2}$; it only affects the choice of p.
REPET	Period index	Reference returns index of $\max + p_{\min}$, i.e. lag +1	Exact lag of the maximum; the search includes $\lfloor m/3 \rfloor$.
REPET	Short fragment ($3p_{\min} > m$)	Empty range in the reference	Search the whole valid range; warn if $r < 3$.
REPET	b normalised by $b(0)$	Yes	Same; only affects the plot.
SIM	Peak picking	`findpeaks` (toolbox)	Own re-implementation with the same rules.
SIM	Frame without peaks	Median of an empty set	$J_j = \{j\} \Rightarrow M = 1$ (nothing extracted).

9 Verification

The tests in `tests/` use synthetic signals with known ground truth and are run headless with `matlab -batch "addpath('core','tests'); run_tests"`.

- `test_hpss`: three sustained sines (440, 880, 1320 Hz) plus a click every 0.5 s, stereo with different gains, at the paper's setting (44.1 kHz, 4096/1024). Checks that the lengths are

17×17 at 44.1 kHz and 19×15 at 48 kHz; that M_H is single, $n \times m \times C$, with values in $\{0, 1\}$; that the mean of M_H is > 0.95 on the sine bins and < 0.05 on the click frames (2–8 kHz) in both channels; and that a silent channel gives $M_H = 0$ with sustained + percussive $= 1$ exactly.

- `test_repet_original`: a 2.0 s loop repeated exactly, plus five short non-repeating notes in their own bands, for 20 s and 19.3 s (non-integer number of periods), mono and stereo. Checks that the detected period is 2.0 s (or an integer multiple) within one hop, mask size and range, mean mask > 0.9 on the loop and < 0.3 on the notes in every channel, the short-fragment case ($r < 3$), and that the four combinations add up to 1 (error $< 10^{-5}$).
- `test_componentes`: 5 s of a real song with the default settings and both repetition methods. Checks that the four combinations add up to 1 and that applying all four components at 0 dB returns the original signal (error $< 10^{-6}$).

These tests verify the implementation against its own specification and the papers' equations; they do not measure separation quality on real music, which must be assessed in the report (listening, spectrograms and the energy measurements).

10 Open issues

1. **Soft HPSS mask** (eqs. 10–11, $p = 1, 2$): the paper's recommended setting is not yet available. Comparing binary and soft masks is direct material for the discussion of side effects.
2. **Mixed mask types.** The voice component $M_H(1-M_R)$ combines a binary and a soft mask: the soft REPET weighting is switched fully on or off by HPSS, so HPSS artefacts dominate at large gains.
3. **Low-frequency behaviour.** With the 100 Hz cut-off removed and the HPSS resolution limit below ≈ 95 Hz, the bass is the least reliable region of every component; restricting the band of a component is the explicit, reportable fix.
4. **Fragment length.** On 20 s fragments k is inactive in REPET-SIM and $p_{\max}$ is capped at ≈ 6.7 s in REPET; both filters were designed for full tracks. This should be stated in the report, or full songs should be analysed and the 20 s excerpt cut afterwards.
5. **Adaptive REPET** (§14.3) is not implemented.

References

[1] D. FitzGerald, "Harmonic/percussive separation using median filtering," in *Proc. 13th Int. Conf. on Digital Audio Effects (DAFx-10)*, Graz, Austria, 2010.

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[3] Z. Rafii and B. Pardo, "REpeating Pattern Extraction Technique (REPET): A simple method for music/voice separation," *IEEE Trans. Audio, Speech, and Language Processing*, vol. 21, no. 1, pp. 71–82, 2013.

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[5] D. W. Griffin and J. S. Lim, "Signal estimation from modified short-time Fourier transform," *IEEE Trans. Acoustics, Speech, and Signal Processing*, vol. 32, no. 2, pp. 236–243, 1984.